

ABHINAV KAUSHIK

Senior Applied Scientist - Demand Forecast

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SUMMARY

Senior AI/ML Engineer with 8+ years of experience across Telecom, AIOps, and Healthcare, with hands-on experience in time-series and demand forecasting using both statistical methods and machine learning techniques such as Holt-Winters, LSTM, and XGBoost. Owns models across the full lifecycle, from prototyping to production deployment with FastAPI and MLflow, working daily in Python, SQL, and PySpark. Experienced in client-facing consulting, translating business requirements into practical, production-ready solutions. Currently building ThinkTree.AI, a non-profit AI-powered learning platform, and looking for opportunities in Berlin.

PROFESSIONAL EXPERIENCE

Senior AI Engineer | Optum Bengaluru, India | Nov 2025 - Present

- Built and integrated a lab extraction agent with an existing Laboratory Information Management System (LIMS) using Claude Opus 4.7 and Amazon Textract, automating work associated with 800 Jira tickets and PRD tasks.
- Engineered a LangGraph-based analytics system for AARP medical plans, combining stateful multi-agent reasoning with real-time chart generation and reducing manual analysis effort by 98%.
- Built automated evaluation and observability pipelines for Question Answering and Summary Agents, leveraging LLM evaluators, quality metrics, tracing, and failure detection to reduce human-in-the-loop review.
- Enhanced dense retrieval performance by training and evaluating a cross-encoder with positive and negative examples for AARP survey data, achieving a 20% increase in retrieval accuracy.

Senior Machine Learning Engineer | Viavi Solutions Chennai, India | Sep 2024 - Nov 2025

- Built 30+ AI application prototypes and mockups for clients across AIOps and GeoPerformance, translating client requirements into functional POCs for solution validation and demonstrations.
- Developed an AI-powered RAG-based Project Discovery Assistant using LangGraph, combining semantic retrieval and LLM reasoning to achieve a 90% match rate with existing organizational solutions.
- Developed time-series forecasting of base-station call handover and SMS handover volumes using LSTM and XGBoost.
- Developed a forecasting feature for the AIOps monitoring product, forecasting univariate KPIs across the network.

Data Scientist | Viavi Solutions Bengaluru, India | Nov 2021 - Sep 2024

- Applied Sentence-BERT and DBSCAN to enhance semantic understanding and cluster alarm messages across graph-based AIOps infrastructure, achieving a 73.5% improvement in semantic alarm classification.
- Applied Isolation Forest and Holt-Winters for anomaly detection and demand forecasting across 500+ telecommunications network devices using 15+ KPIs, supporting earlier identification of abnormal time-series behavior and proactive network planning.
- Delivered technical POCs and consulting sessions on data services and products for 15+ clients across 3+ AIOps and ML platforms, contributing to an 11% increase in POC and mentoring revenue.

Data Analyst | United Health Group Gurugram, India | Jun 2021 - Nov 2021

- Collaborated with stakeholders on ad-hoc analysis and customer-retention dashboards using Hive, SQL, and Power BI.
- Developed a Python- and Flask-based parser to automate data extraction, parsing, and transformation into structured data, supporting downstream analytics and reporting.

Data Engineer | Nippon Data Systems Gurugram, India | Nov 2019 - May 2021

- Engineered SQL and PySpark-based data pipelines to automate BAU medical-audit reporting, transforming and aggregating large healthcare datasets and reducing manual reporting effort by 95%.
- Built real-time observability using Elasticsearch for centralized log ingestion and analysis, with Kibana dashboards for monitoring system activity and user events, improving traceability efficiency by 80% for the COVID QRT team.
- Developed a Flask-based enterprise search engine using Apache Solr and Lucene indexing, enabling centralized search and retrieval across organizational data for enterprise-wide use cases.

Associate Consultant | Saksoft Gurugram, India | Dec 2017 - Nov 2019

- Conceptualized customer dashboards using QlikSense, SQL, and SAP HANA for financial and product insights, resulting in a 40% increase in user engagement.
- Developed an XGBoost-based propensity model to predict the probability of customers initiating a personal loan application within 7 days of contact, enabling targeted outreach and customer prioritization.
- Leveraged Selenium and Python to automate performance testing and evaluate application response times, resulting in a 90% reduction in testing effort.

PROJECTS

Think Tree | ML Study Platform

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.

Retail Demand Forecasting Pipeline

- Built a hierarchical demand forecasting pipeline for retail product-level sales data, combining XGBoost and LSTM models to capture cross-product and temporal patterns.
- Evaluated forecast accuracy against statistical baseline methods and served predictions through a FastAPI service with MLflow for model tracking and versioning.
- **Technologies:** Python, SQL, PySpark, XGBoost, LSTM, FastAPI, MLflow, Pandas

Financial Fraud Detection | Mar 2025

- Developed an AI-driven fraud detection and reasoning workflow using the JPMorgan Synthetic Dataset, combining LightGBM-based modeling with LLM explanations to support interpretable financial-risk analysis.

Restaurant Feedback Analyzer | Aspect-Based Sentiment Analysis

- Built an NLP pipeline to extract review aspects and determine sentiment at the aspect level, enabling granular insights into customer experience.

RFP Multi-Agent Proposal Automation Platform

- Built a multi-agent RFP automation platform using an LLM-driven supervisor to orchestrate intent detection, retrieval, proposal generation, document matching, and market research with RAG over historical proposals.

EDUCATION

Master of Science, Machine Learning and AI Liverpool John Moores University	Jan 2021 - Jan 2023
Bachelor of Technology, Information Technology Jamia Hamdard	Aug 2013 - Aug 2017

TECHNICAL SKILLS

AI / GenAI: GenAI, LLM, RAG, Agentic AI, LangChain, LangGraph, Claude, OpenAI, Embeddings, Tool Orchestration, Cross-Encoder, Transformers, NLP, Generative AI
Machine Learning: XGBoost, LightGBM, Random Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Statistical Modeling, Regression, Deep Learning
Programming / Data: Python, SQL, PySpark, Pandas, NumPy, Flask, Hive, SparkQL, REST API, Pydantic, Unix
MLOps / Cloud: MLflow, Docker, Kubernetes, Jenkins, Bamboo, Bitbucket, DVC, Terraform, Airflow, GCP, Azure, NCI Servers
Search / Observability: Elasticsearch, Kibana, Logging, Log Analysis, Apache Solr, Lucene Indexing, Search and Information Retrieval, AI Evaluation, LLM Evaluators, Agent Tracing
Data Platforms: Databricks, Apache Spark, Airflow
Databases: TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB
BI / Testing: Power BI, QlikSense, SAP HANA, Selenium

LANGUAGES

English: Native
German: A1, Pursuing

ADDITIONAL INFORMATION

- Network Intern, Airport Authority of India
- Hadoop Intern, Ducat Technologies